



Decimal BS

Normal BS

Transforming Equations for Binary Search

Monotonic Fn

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Problem 1



Pair Selection -

<https://codeforces.com/edu/course/2/lesson/6/4/practice/contest/285069/problem/C>

$$a = \{a_1, a_2, \dots, a_n\}$$

$$b = \{b_1, b_2, \dots, b_n\}$$

$$k = 2$$

$$(a_i / b_i)$$

$$\left. \begin{array}{cc} a_l & a_r \\ b_l & b_r \end{array} \right\}$$



sort

$$\begin{array}{ccc}
 \downarrow & & \downarrow \quad \downarrow \\
 \frac{a_l + a_r}{b_l + b_r} & \neq & \frac{a_l}{b_l} + \frac{a_r}{b_r}
 \end{array}$$

A large curly brace spans the entire expression above. A smaller curly brace is positioned under the right-hand side of the equation, specifically under the sum of the two fractions.

BS on a ratio x

①

$\neg$ If there exists a choice with
ratio $\geq x$

then also exists a choice with
ratio $\geq x$ - something

✓

Monotonic Predicate Fn

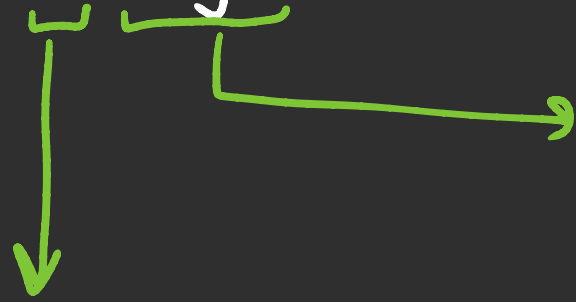
$f(x) \rightarrow \begin{cases} T & \text{if there exists a ratio } \geq x \\ F & \text{otherwise} \end{cases}$

low

T T T T T T T T F F F F F F

high

$O(n \log n)$



BS

while
for (it) ... } $O(\log ss)$

checker fn



$O(n)$ →

Was a ratio there for some
choice of k points which is
 $\geq X$

$$\Rightarrow \frac{a_1 + a_2 + a_3 \dots a_k}{(b_1 + b_2 + b_3 \dots b_k)} \geq X$$

$$\Rightarrow a_1 + a_2 + a_3 \dots a_k \geq X(b_1 + b_2 + b_3 \dots b_k)$$

$$\Rightarrow (a_1 + a_2 \dots a_k) - X(b_1 + b_2 \dots b_k) \geq 0$$

$$(a_1 - x b_1) + (a_2 - x b_2)$$

$$\dots\dots (a_k - x b_k) \geq 0$$

$$(a_1, b_1) (a_2, b_2) \dots\dots (a_n, b_n)$$

Transformed

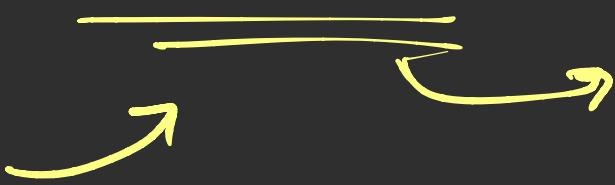
$$C_i = A_i - x B_i$$



$$C_1 + C_2 + C_3 + \dots C_k \geq 0$$

↳ Take up k elements st sum is ≥ 0

Sort this



and then
take best k top
elements

for () $\leftarrow$ BS $O(\log ss)$

{ mid (x)

$\rightarrow$ TA $\rightarrow$ $O(n)$
 $\downarrow$ $\downarrow$ $O(n \log n)$
checker ($O(n)$)

{ sort
for }

}

$$O(\log_2(ss) [n + n \log_2 n + n])$$

$$O(\log_2(ss) \times n \log_2 n)$$

$$\approx O(n(\log_2 n)^2)$$

k pairs

5 x

6

7

8

9

Problem 2



Average & Median [Hard] -

https://atcoder.jp/contests/abc236/tasks/abc236_e

N

$a = \{ a_1 \quad a_2 \quad a_3 \quad a_4 \quad a_5 \quad a_6 \}$

↳ if you take some elements, the
gap between them ≤ 1

BS on the avg (x)

$f(x) \rightarrow T$ - if $\text{avg} \geq x$

$\rightarrow F$ - otherwise

T T T T T T T T F F F F

$O(n \log(ss)) \rightarrow \underline{\underline{BS}}$

$\rightarrow$ checker fn $\rightarrow$

was a avg $\geq X$ possible

C = Count of elements I took

$$\downarrow \frac{a_1 + a_2 + a_3 \dots a_C}{C} \geq X$$


$$a_1 + a_2 + a_3 \dots a_C \geq X C$$

$$\overbrace{a_1 + a_2 + a_3 \dots a_c} - \overset{\downarrow}{X} \overset{\swarrow}{C} \geq 0$$

$$(a_1 - X) + (a_2 - X) + (a_3 - X) \\ \dots \geq 0$$

Transformed Arr = C

$$\Rightarrow \underline{c_i = a_i - x}$$

$$\{c_1 \quad c_2 \quad c_3 \quad c_4 \quad \dots \quad c_n\}$$



$$\geq 0$$



55
56 which tells me whether a median $\geq m$ is possible or not - TRUE/FALSE

57
58 Let's say that I created a new modified array c

59
60 where $c[i] = 1$ if $a[i] \geq m$
61 else $c[i] = -1$ if $a[i] < m$

62
63 If I run my previous logic of finding the maximum sum thing (d array) on this new array c
64 Can I say I want now the same thing, that is the maximum sum > 0

65
66 why? because I have greater number of 1s compared to -1s
67 so my sum of the chosen elements from array c should give me a sum > 0

```
31  ans<=0 can you say, you have some avg >= x - TRUE
32  else FALSE
33
34
35  for some number to be a median? what has to satisfy???
36
37  let's say I have a array of size n
38
39  arr = {arr1, arr2, arr3, arr4, arr5}
40
41  median - m
42
43  the number of elements that are >=m should be greater
44  than the number of elements that are less than m???
45
46  We want to binary search on the median.
47
48  Assuming we have a median - m
49
50  if I don't find a median >= m, then I cannot find a median >= m+1, m+2, m+3, .....
51
52  TTTTTTTTTTTT....FFFFFFFFF
53
54  I want a checker fn -
55
```

```

3  c[i] = a[i]-X
4
5  I have an array d of size n
6
7  where I define the following,
8
9
10 d[i] = maximum sum of elements from 0 to ith index
11 if I have always chosen c[i] / always picked up ith index
12
13 c = {2, -1, 4, -5}
14
15 d[0] = 2
16 d[1] = -1 + max(d[0], 0) = 1
17 d[2] = 4 + max(d[1], d[0]) = 6
18 d[3] = -5 + max(d[2], d[1]) = 1
19
20 d[i] = c[i] + max(d[i-1], d[i-2])
21
22 ans = max(d[n-1], d[n-2]) = maximum sum possible by choosing elements in such a way
23 such that no two chosen elements have a gap > 1
24
25 d[n-1] = maximum sum of elements from 0 to (n-1)th index
26 if I have always chosen c[n-1]
27
28 d[n-2] = maximum sum of elements from 0 to (n-2)th index
29 if I have always chosen c[n-2]

```

Problem 3



Max Average Segment [Homework] -

<https://codeforces.com/edu/course/2/lesson/6/4/practice/contest/285069/problem/A>